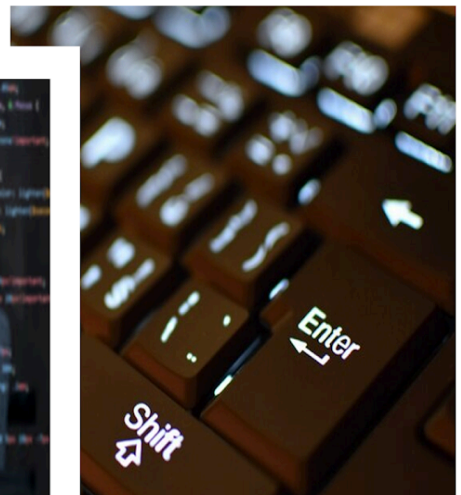


Computer Awareness

STUDY NOTES

Number System



Number System

The number system is a method of Counting.

General Representation

$(N)_b$

Where, N = No. to be represented

b = base or radix

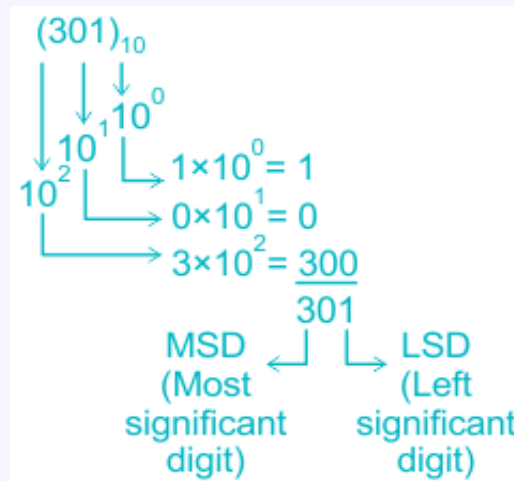
Range = (0 to b - 1) i.e. N can be b/w (0 to b - 1)

Commonly used Number system

- Decimal Number System
- Binary Number System
- Octal Number System
- Hexadecimal Number System

Decimal No. System

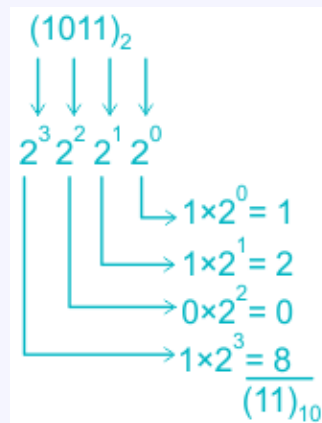
- Base i.e. 'b' is 10
- $(N)_{10}$ for eg. $(301)_{10}$
- Range (0 - 9) i.e. N can contain digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)
- Value of digit depends upon its position



LSD is least Significant Bit.

Binary Number System

- Base i.e. 'b' is 2
- $(N)_2$ i.e. $(1011)_2$
- Range (0 - 1) i.e. N can contain only (0, 1)
- This 0, 1 in Binary system is known as bits
- Group of 4 bits = 1 Nibble
Group of 8 bits = 1 byte
- Follows positional value system i.e. Binary number can be converted to Decimal equivalent by summery the weight of various position in binary number which contain bit '1'



Octal Number System

- Base i.e. 'b' is 8
- $(N)_8$ i.e. $(367)_8$
- Range (0 - 7) i.e. N can contain digits (0, 1, 2, 3, 4, 5, 6, 7)
- Octal Number is converted to Decimal equivalent by summery together the positional weight of each octal digit.

$$\begin{array}{r}
 (367)_8 \\
 \downarrow \downarrow \downarrow \\
 8^2 \quad 8^1 \quad 8^0 \\
 \downarrow \quad \downarrow \quad \downarrow \\
 3 \times 8^2 = 192 \\
 6 \times 8^1 = 48 \\
 7 \times 8^0 = 7 \\
 \hline
 (247)_{10}
 \end{array}$$

Hexadecimal Number System

- Base i.e. 'b' is 16



- Also known as Alpha Numeric Number system
- Range (0 - 15)

$$\begin{array}{r}
 (E5)_{16} \\
 \downarrow \downarrow \\
 16^1 \quad 16^0 \\
 \downarrow \quad \downarrow \\
 E \times 16^1 = 14 \times 16 = 224 \\
 5 \times 16^0 = 5 \\
 \hline
 = (229)_{10}
 \end{array}$$

Number System Conversion

Decimal to any Number system

Decimal Number

- **Integer part** - Divide the integer part by 'base' {2, 16, 8} & write the remainder until quotient obtained is zero.
- **Fractional part** - Multiply the fraction part by 'base' {2, 16, 8} & write the integer part till we get fraction equal to zero or we get the repeated pattern.

Decimal to Binary

$$(43.90625)_{10} \rightarrow (101011 \cdot 111010)_2$$

	Remainder
43 ÷ 2 → 1	(LSB)
21 ÷ 2 → 1	↑
10 ÷ 2 → 0	
5 ÷ 2 → 1	
2 ÷ 2 → 0	
1 ÷ 2 → 1	(MSB)
0.90625 × 2 = 1.8125	
0.8125 × 2 = 1.625	
0.625 × 2 = 1.25	
0.25 × 2 = 0.5	
0.5 × 2 = 1.0	
0.0 × 2 = 0.0	
Now fraction become '0'	
$(43)_{10} \rightarrow (101011)_2$	

	'Integer'
0.90625 × 2 = 1.8125	↓
0.8125 × 2 = 1.625	
0.625 × 2 = 1.25	
0.25 × 2 = 0.5	
0.5 × 2 = 1.0	
0.0 × 2 = 0.0	
Now fraction become '0'	
$(0.90625)_{10} \rightarrow (0.111010)_2$	

Decimal to octal

$$(43.30)_{10} \rightarrow (53.23146)_8$$

Reminder

$$43 \div 8 \rightarrow 5 \text{ remainder } 3 \quad 5 \div 8 \rightarrow 0 \text{ remainder } 5 \quad \uparrow \quad (43)_{10} \rightarrow (53)_8$$

$$\begin{array}{llll}
 0.30 \times 8 = & \rightarrow & \downarrow (0.30)_{10} \rightarrow & \\
 2.40 & 2 & (0.23146)_8 & \\
 0.40 \times 8 = & \rightarrow & & \\
 3.20 & 3 & & \\
 0.20 \times 8 = & \rightarrow & & \\
 1.60 & 1 & & \\
 0.60 \times 8 = & \rightarrow & \text{Repeat} & \\
 4.80 & 4 & & \\
 0.80 \times 8 = & \rightarrow & & \\
 6.40 & 6 & & \\
 0.40 \times 8 = & \rightarrow & & \\
 3.20 & & &
 \end{array}$$

Any Number system to Decimal Number system

Binary to Decimal

$$(1101.01)_2 \nearrow 1101 \rightarrow 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13 \quad 01 \rightarrow 0 \times 2^{-1} + 1 \times 2^{-2} = 0 + \frac{1}{4} = 0.25$$

$$(1101.01)_2 \Rightarrow 13 + 0.25 = (13.25)_{10}$$

Octal to Decimal

$$(46.2)_8 \nearrow 46 \rightarrow 4 \times 8^1 + 6 \times 8^0 = 38 \quad 0.2 \rightarrow 2 \times 8^{-1} = 0.25$$

$$(46.2)_8 \Rightarrow 38 + 0.25 = (38.25)_{10}$$

Hexadecimal to Decimal

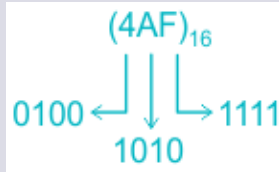
$$(E7.2)_{16} \nearrow E7 = E \times 16^1 + 7 \times 16^0 = 231 \quad 0.2 = 2 \times 16^{-1} = \frac{2}{16} = \frac{1}{8} = 0.125$$

$$= 231 + 0.125$$

$$= (231.125)_{10}$$

Hexadecimal to Binary

Replace hexadecimal Number by its 4 bit binary equivalent.



$$(4AF)_{16} \rightarrow (0100\ 1010\ 1111)_2$$

Octal to Binary

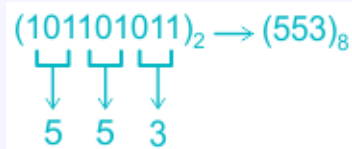
→ Replace Octal Number by its 3 bit binary equivalent

$$(376)_8 \rightarrow (011\ 111\ 110)_2$$



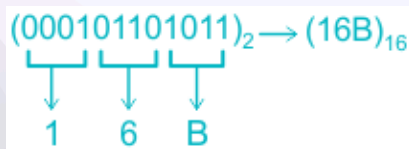
Binary to Octal

Group the binary bit stream in to pairs of '3' by starting it from LSB



Binary to Hexadecimal

Group the binary bit stream into pairs of '4' by starting it from LSB



Octal to Hexadecimal

→ Convert octal to Binary number

→ Perform Binary to Hexadecimal conversion

$(553)_8 \downarrow (101\ 101\ 011)_2 \downarrow$ Binary equivalent Binary to Hex

101101011

 $(16B)_{16}$

Hexadecimal to Octal

→ Convert Hexadecimal to Binary Number

→ Perform Binary to octal conversion

$(16B)_{16} \downarrow (0001\ 0110\ 1011)_2 \downarrow$ Binary equivalent Binary to Octal

000101101011

 $(553)_8$

Signal Number Representation

This is useful for the representation of negative Number

Mostly used are: -

→ Signal Magnitude Representation

→ 1^s complement representation

→ 2^s complement Representation

Signed Magnitude	1^s Complement	2^{nd} Complement
1) MSB is a sign bit For -ve no., sign bit = 1 For +ve no., sign bit = 0	-ve no. is represented in 1^s complement form	+ve no. is represented in 2st complement form.
2) Two pattern for '0'	Two pattern for '0'	Single pattern for '0'

3) Use different hardware for addition and subtraction	Use same hardware for addition and subtraction	Use same hardware for addition and subtraction
4) Weighted No. System	Non weighted	Weighted no. system
5) Do not assign weight to sign bit	-	Assign weight to sign bit i.e., -ve weight to sign bit +ve weight to rest bits
6) $-2^{n-1} - 1$ to $+2^{n-1} - 1$	-2^{n-1} to $+2^{n-1} - 1$	-2^{n-1} to $+2^{n-1} - 1$
7) Sign bit does not involve in computation	Sign bit involve in computation	Sign bit involve in computation.

Testbook Trick:

For signed number extension

Signed Magnitude	1 st and 2 nd complement
→ put '0' b/w sign bit and successive bit	→ copy 'sign bit' b/w sign bit and successive bit

Codes

Codes		
Non Binary	Binary	alphanumeric
Morse code	→ Gray code → Excess 3 code → BCD code → 8421 code	→ ASCII code → EBCDIC code

Weighted Codes

- 8421
- BCD
- Binary, Hex, oct

Non weighted codes

- excess 3
- Gray code

Self-complementary codes

→ They are weighted code with sum of weight = 9 and They have property that 9's complement of decimal no. is obtained directly by complementing each bit in the pattern.

Eg. 2421, excess 3, 8 4 -2 -1, 5211, 4311, 3321

Testbook Trick:

Excess 3 is the only Non-weighted code which is self-complementing.

BCD code (Binary code Decimal)

- Each digit of decimal Number is represented by binary equivalent
- Also known as 8421 code or 4 bit binary code.
- during arithmetic operation of invalid BCD is present then add 0110 to get correct result

$(46)_{10} \text{ BCD} \rightarrow 0100 \ 0110_{\text{BCD}}$

Excess 3 code

→ $\text{BCD} + 3 = \text{Excess 3 code}$

For e.g.

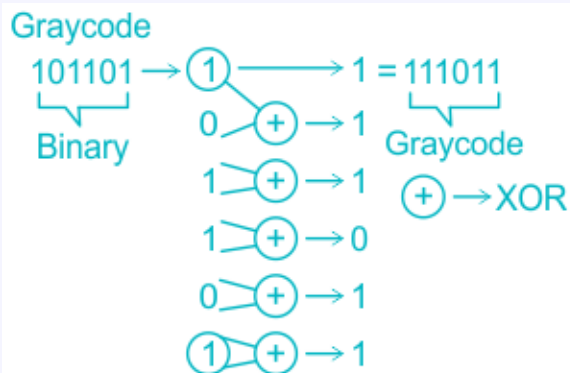
$(43)_{10} \longrightarrow$ 0100 0011 $\rightarrow$ BCD
 Excess 3 code 0011 0011 $\rightarrow$ 3
 0111 0110 $\rightarrow$ Excess 3 code

Gray code

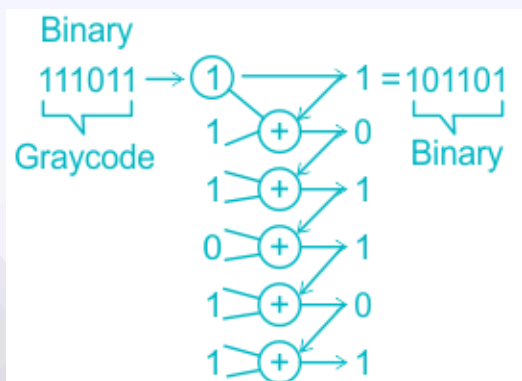
$\rightarrow$ Also known as cyclic code, Minimum error code, Reflective code.

$\rightarrow$ In this code, only one bit in the code group changes when going from one step to next.

Binary to Gray code



Gray code to Binary



Points to Remember

- Gray codes are not suitable for arithmetic operations.
- Excess 3 code arithmetic operation used to perform BCD subtraction
- Largest no. that can be represented using N bits = $(2^N - 1)_{10}$
- BCD code is used in counters, Digit clocks etc.

ASCII Code (American Standard Code for Information Interchange)

- A 7-bit character encoding standard for text representation in computers.
- **Character Set:** Supports 128 characters, including control characters (0-31) and printable characters (32-126).
- **Limitations:** Primarily supports English; insufficient for other languages.

Extended ASCII Code

- An 8-bit extension of ASCII, allowing for 256 characters.
- **Character Set:** Includes additional symbols, accented letters, and graphics characters.
- **Variations:** Different sets (e.g., ISO 8859-1, Windows-1252) cater to various languages.

ISCII Code (Indian Script Code for Information Interchange)

- An encoding scheme for Indian scripts developed by the Government of India.
- **Character Set:** Supports multiple Indian languages like Hindi, Bengali, Tamil, etc.
- **Features:** Uses variable-length encoding and includes representations for vowels, consonants, and diacritics.